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## TOPICAL REVIEW

# Determine the Exact Fault Location in Underground Cable: A Systematic Review

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**ABSTRACT** In urban areas, the power distribution system is based on medium-voltage energy cables. Typically, place cables underground in urban; finding faults in cables is very challenging. This paper reviews several fault detections and location identifications. In addition, the paper relies on the study of cable. Existing published articles play a vital role in trying to target cable durability, reliability, cable modeling, protection relay, and monitoring the cable voltage and current in the substation. The underground cable is not affected by thunder, climate change, rain, or snow. Due to aging effects, cable sheathing can occur. This leads to incipient and arcing faults. The possible fault will fail to open and short-circuit. This survey explains cable modeling, faults, fault detection techniques, estimates of fault location of various trees, protection, and electrical and water trees. Finally, the results of the stimulation metrics are displayed.

**INDEX TERMS** Underground cable, fault, wavelet, time domain reflectometry, Arduino, transient.

## I. INTRODUCTION

In urban areas, the demand for power supplies increases due to increasing population, and the transmission of electricity is more difficult to place in overhead transmission and distribution [1], [2]. So, cables are widely used in medium voltage distribution and are reliable for uninterrupted power supply. In the case of a fault in a UG cable, it is significantly hard to locate the fault location in the underground cable due to its concealed nature, making fault detection and location difficult, because the cables are not visibility accessible after installation [3], [4]. This review explains the various types of generation, transmission, and distribution losses for medium voltage circuits. Medium-voltage circuit designed according to IEEE-13node feeder figure: 1, 2 underground cable designed according to the IEC-60502-2 and IEC 60228 standards [5], [6], [7]. In this survey, explain the specific significance of voltage, current, cable dimensions, construction, testing, and requirements for the medium voltage distribution (10 kV to 30 kV) [8], [9], [10], [11]. In this standard, failure of the cable sheathing is explained. In a rainy situation, water penetrates the cable. So, the reason for loss or fault may be the occurrence of energy affected

by water trees and electrical trees due to aging or physical damage. In the fault, the water tree is also considered the electrical tree [12], [13]. In this tree, also consider a similar calculation of the offline incipient fault at the arc. In medium-voltage underground cables, techniques have been able to detect the occurrence of faults at any location. In this paper, explain Ohm's law with Arduino, Fast Fourier, DWT, and sliding window submission fault location in underground cable findings in monitoring systems at substations by using IoT and cloud computing.

## A. MOTIVATION

In urban power networks, electrical energy is predominantly distributed through underground cables due to their improved reliability and uninterrupted supply compared with overhead lines [3], [14], [15]. However, underground cable systems remain vulnerable to various types of faults, including insulation breakdown, sheath failure, incipient arcing, open-circuit faults, and short circuits. Identifying such faults is challenging, and manual detection methods are often time-consuming and labor-intensive [16], [17]. Fault localization typically requires extensive excavation and field inspections, which result in prolonged downtime, increased maintenance costs, and significant disruption to public infrastructure.

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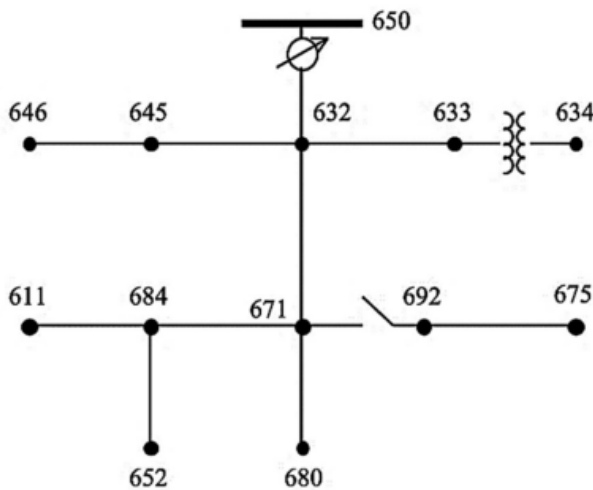


FIGURE 1. IEEE 13 node feeder tester [159].

Therefore, there is a strong requirement for fast, accurate, and automated fault-location techniques to ensure reliable system operation [18], [19], [20]. Recent advancements have encouraged the development of intelligent fault-location systems that leverage modern analytical approaches, including machine learning and deep learning models, Ohm's law-based estimation, frequency-domain analysis, wavelet transforms, sliding-window processing, and resonance-based algorithm.

## B. CONTRIBUTION

The major findings of the review on underground cable fault location techniques are summarized as follows. A systematic investigation of existing methodologies for fault detection and localization has been conducted. The reviewed approaches have been analyzed based on the underlying techniques, datasets utilized, and evaluation metrics reported. In addition, this survey has provided an integrated perspective of the research landscape by identifying limitations, highlighting unresolved challenges, and outlining potential directions to enhance existing frameworks and methodologies. The structure of this paper is organized as follows:

- **Section I** provides the introduction, motivation, contributions, and search methodology used in this review.
- **Sections II and III** describe the available datasets, cable signal processing techniques, and commonly used performance metrics.
- **Sections IV and V** present the methods for fault classification and localization reported in the literature, followed by the overall discussion and conclusion.
- **Sections VI and VII** outline unresolved research challenges and propose potential future research directions toward intelligent underground cable fault detection systems.

## II. SEARCH STRATEGY

The searched articles have been collected from IEEE Xplore, Scopus, Web of Science, ScienceDirect, PubMed, and

Google Scholar for publications appearing between January 2014 and March 2025 on the application of fault location in underground cables. The keywords used to download the papers have included the study topic, conference papers, review articles, pediatric journals, eBooks, conference paper downloads, and selected articles from various databases table 7.

The initial search has yielded a large number of articles, including duplicates published under similar authors, keywords, or titles. During the screening process, articles with low relevance, those that have not provided clear solutions, and those that have excluded underground-cable-specific criteria have been removed. Priority has been given to highly relevant publications, including Q1 journals, to reduce human bias in the selection process.

### A. INCLUSION CRITERIA

- Studies that have focused on cable modeling, fault types, and accurate fault-location detection in underground cables have been included.
- Research that has examined water-tree degradation has been included.
- Studies that have focused on transient and arcing faults have been included.
- Q1 journal publications, Review articles, and Conference papers have been included.

### B. EXCLUSION CRITERIA

- Studies related to transmission lines have been excluded.
- Research focused on water-tree degradation has not been considered.
- Theses have not been considered.

### C. IDENTIFICATION

1) TOTAL ARTICLES DOWNLOADED: 303

2) DUPLICATE REMOVAL

- Excluded: 47 duplicates
- Remaining after duplicates removed: 256
- Excluded (based on title/abstract relevance): 46
- Remaining: 210

3) RELEVANCE SCREENING

- Excluded (based on Q1selection): 41
- Remaining (Q1 Articles): 169

### D. DISCUSSION

This review has evaluated the state-of-the-art methodologies in underground cable fault detection, including machine learning (ML) and deep learning (DL) techniques, Ohm's law-based frameworks, and incipient fault analysis methods. The assessment encompassed feature extraction strategies, classification algorithms, and hybrid architectures integrating ML and DL models. The systematic review was conducted using more than 200 publications spanning the period from 2013 to 2025.

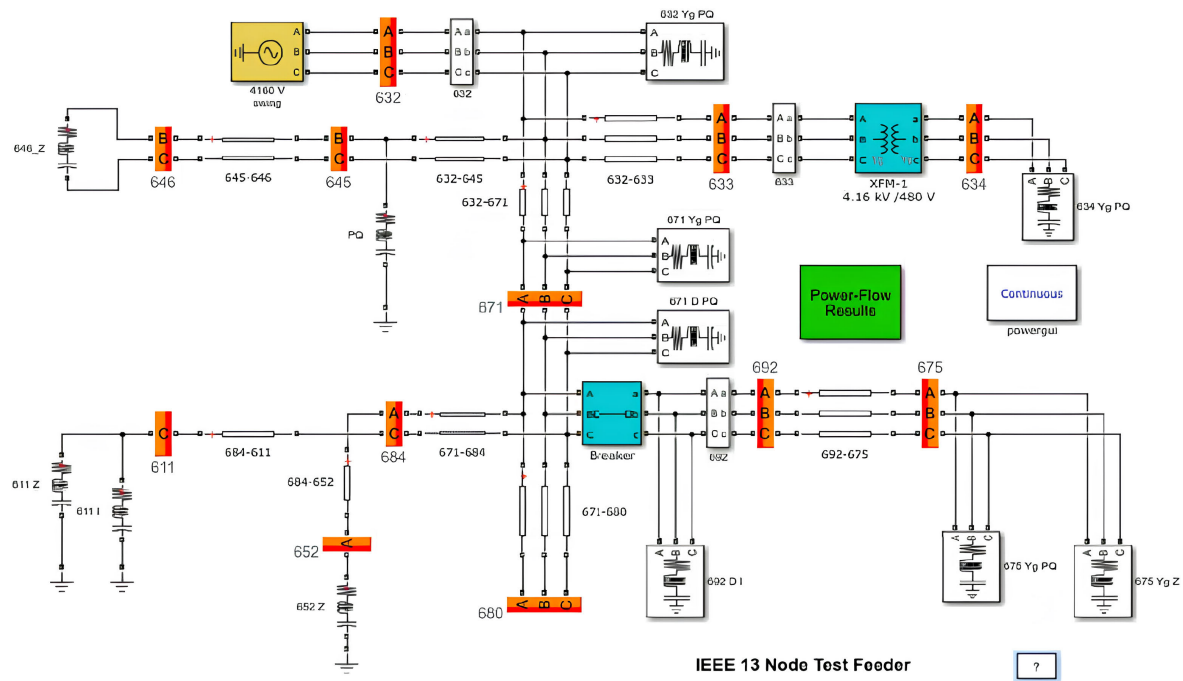


FIGURE 2. Simulink block diagram of the distribution model [158].

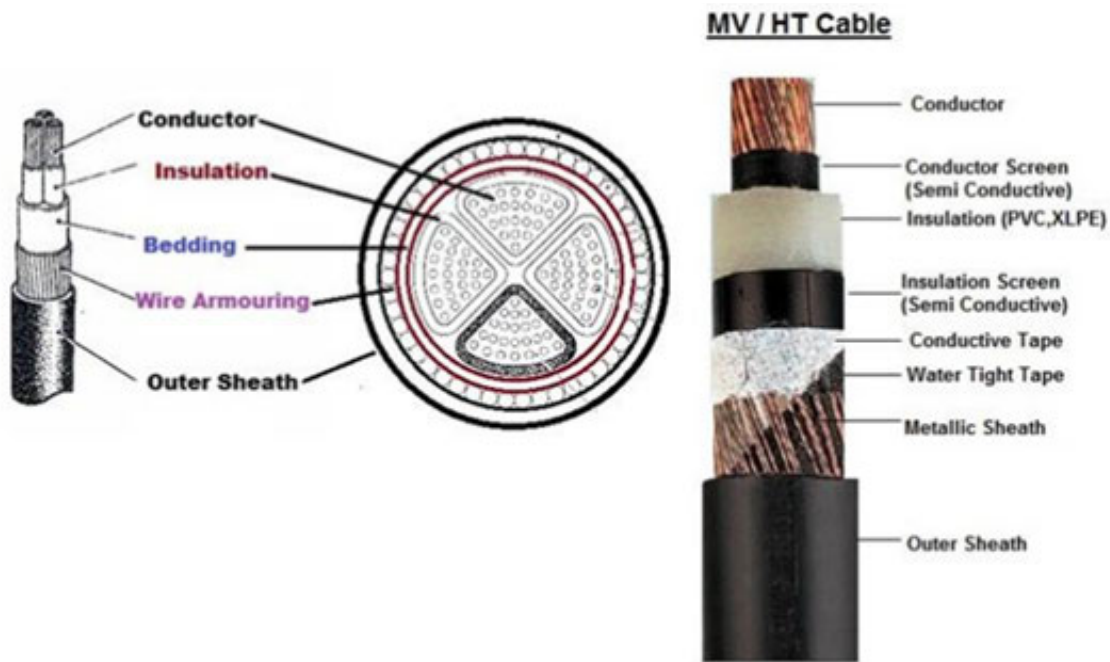


FIGURE 3. Parts of an underground power cable [162].

Based on the findings, several potential research pathways have been identified, particularly those offering strong feasibility for real-time and field-deployable applications.

E. PROPOSED TECHNIQUES AND METHODS

Various methods are used to determine the exact fault location in an underground cable. Results are summarized using accuracy, sensitivity, specificity, etc..

III. CABLE

An underground cable contains one or more conductors surrounded by an appropriate dielectric medium. Its major components, including the insulation, sheath, bedding, armor, and serving, are illustrated in the figure: 3 [91], [92], [93], [94].

The application of an underground cable also depends on the choice of conductor, which is selected based on

**TABLE 1. Travelling waves-based underground cable fault detection summary table.**

| S.No | Citation | Introduction                                                                            | Methodology                                                      | Conclusion                                                          |
|------|----------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------------------|
| 1    | [128]    | DC voltage applied for fault identification based on Ohm's law using a microcontroller. | Microcontroller-based with relay, ADC, LCD                       | Fault identified using Ohm's law; future enhancement with GSM       |
| 2    | [129]    | Varely loop method used for long-distance fault detection.                              | Varely loop, Wheatstone bridge, ADC                              | Accurate location using Arduino and Varely loop method              |
| 3    | [130]    | Ohm's law with PIC and ESP8266 microcontroller for fault detection.                     | Fault types explained (LG, LL, LLG, LLL), Proteus and AVR Studio | Fault detection achieved using IoT and Ohm's law                    |
| 4    | [131]    | Underground cable faults identified using Arduino and Ohm's law.                        | Ohm's law-based voltage drop via resistor                        | Location detected via Arduino; both short & open circuit targeted   |
| 5    | [132]    | Faults like open/short circuits are located with Arduino & Ohm's law.                   | Online/Offline detection, Tracer, Terminal, Murray Loop          | Accurate pinpointing using ratio method                             |
| 6    | [133]    | Fault location detected via internet-based system using Ohm's law.                      | Online & Offline methods (Tracer, Terminal)                      | Could detect short circuit fault; future scope includes open faults |
| 7    | [134]    | Uses LSTM and resonance frequency for fault location.                                   | LSTM (RNN) + resonance frequency analysis                        | Accurate localization through LSTM and system resonance             |
| 8    | [135]    | Fault detected using Ohm's law and ESP8266-based system.                                | Fault types identified, Ohm's law + IoT (ESP8266, Megger)        | Location found through voltage variation                            |

**TABLE 2. Incipient-based underground cable fault detection summary table.**

| S.No | citation | Introduction                                                                                                                                                                             | Methodology                                                     | Accuracy                                                | Conclusion                                                                                                                          |
|------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 1    | [136]    | ANFIS fuzzy logic and DWT have been used to identify faults in medium-voltage underground cables. The inputs have included energy extracted from three-phase and zero-sequence currents. | Simulated using ATP at 200 kHz; model uses Sugeno FIS.          | Accurately identifies fault type and location.          | Inputs are trained using ANFIS with detail energy to identify fault type and location. Suggests potential for MV cable diagnostics. |
| 2    | [137]    | Fault detection using wavelet MRA and traveling wave theory in high-voltage cables.                                                                                                      | Current passed through LPF HPF filters; MRA applied.            | Achieves precise fault location via wave time response. | Wavelet transform used to detect peak energy time using aerial mode coefficients. Recommends use for extra high voltage systems.    |
| 3    | [138]    | Deep learning approach for fault classification and location in nuclear facility cables.                                                                                                 | Wavelet features extracted and applied to deep neural networks. | Demonstrates high accuracy and reliability.             | Combines DWT with deep neural nets for classification/location. Suggests expansion to real-world power systems.                     |

flexibility, economy, physical properties, shape, voltage level, and other relevant factors [95]. The radius measurements corresponding to various multiples of R value are shown in the figure: 4 and table: 8.

## A. ELECTRICAL CONSIDERATION

### 1) CORE

The core has been made of aluminum or copper, depending on the application. The cable consists of one or more conductors surrounded by insulation, as required for its intended use. The cable voltage has been affected by proximity and skin effects, both of which have been controlled by using a standard conductor [96].

### 2) INSULATOR

An insulator ensures that electrical current flows only through the conductor and does not escape to the surroundings. It was preventing contact between phases or with the neutral/earth. because, the thickness has directly affected the rated voltage. [97].

### 3) METALLIC SHEATH

In cables, the metallic sheath has been made of lead or aluminum, and it has served as a barrier to prevent moisture penetration, which is a major cause of insulation failure or damage due to liquids present in the soil during earth faults [96].

### 4) BEDDING

The cable bedding has consisted of materials used to provide separation from the metallic sheath and to protect the cable from mechanical injury caused by armoring [97], [98].

### 5) ARMORING

The armoring has been provided to ensure good mechanical strength and protection. It has been placed above the bedding layer and typically has been made from high-resistance galvanized steel [98].

- **Advantages:** The cable has exhibited low losses, low maintenance requirements, high reliability, and immunity to climate changes and snow.

TABLE 3. Fault detection techniques using wavelets and neural networks-based underground cable fault detection summary table.

| S.No | Title | Introduction                                                                         | Methodology                                                                                                                  | Accuracy                                                                                                | Conclusion                                                                                     |
|------|-------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| 1    | [143] | HVDC systems have faced challenges in detecting high-resistance faults.              | VMD and DRSN have been used for fault localization, and the DRSN has been trained on decomposed IMF signals from VMD.        | Fault location error has been reported below 1.5 km and performance has remained resilient under noise. | Future work has suggested integrating real-time specifications and expanding noise modeling.   |
| 2    | [144] | Delays in fault detection in power systems have been addressed.                      | DWT with Newton–Raphson optimization has been used, including a two-stage protective relay tuning scheme.                    | The model has detected faults within 0.01 s with high accuracy.                                         | Smart grid integration and topology adaptability have been recommended.                        |
| 3    | [145] | TW propagation in power networks has suffered frequency distortion.                  | A frequency-corrected TW analysis framework has been proposed using GPS synchronized measurements and reflection management. | Propagation speed has been corrected per section, improving localization accuracy.                      | The study has demonstrated improved performance for varied OHTL configurations.                |
| 4    | [146] | Hybrid overhead–underground tree flash faults have been investigated.                | VMD and Hilbert transforms have been applied with a dual-end identification method.                                          | The method has achieved high accuracy in simulation.                                                    | Real-world implementation and environment adaptability have been recommended.                  |
| 5    | [147] | LV systems have been insufficiently represented in IEEE Std. 1234-19.                | A field-tested 8-step fault handling procedure has been proposed with a manual flowchart approach.                           | The method has served as the first LV-specific practical guide.                                         | Future recommendations have included SCADA/PMU integration and full automation.                |
| 6    | [148] | Hybrid line fault localization needs have been identified.                           | IARC has been applied with SOGWO optimization; time difference has been converted into distance.                             | The method has remained accurate under various grid conditions.                                         | Future work has suggested expanding to real-time grids and additional fault scenarios.         |
| 7    | [149] | Low-cost fault localization using existing field devices has been proposed.          | The time constant of an injected signal has been used with single-ended multi-injection analysis.                            | The method has demonstrated robustness across noise levels and cable variations.                        | Implementation in practical LV DC grids has been recommended.                                  |
| 8    | [150] | PMU data have been explored for distribution network fault detection.                | Synchrophasor-based multi-end localization has been used.                                                                    | Accurate short-circuit fault detection has been reported.                                               | Future work has proposed integration with smart meters and improved classification schemes.    |
| 9    | [151] | Speed improvements in underground cable fault classification have been investigated. | Wavelet MRA and ANN have been used, including spike-based classification.                                                    | The approach has shown fast, phase-specific detection capability.                                       | Future enhancements have suggested extension to transient faults and mixed loading conditions. |
| 10   | [152] | Offshore HVDC fault detection challenges have been examined.                         | A Numerical Laplace Transform-based TW approach has been used to convert signals into the frequency domain.                  | The method has provided accurate localization for PTG and PTP faults.                                   | Future directions have included scaling to multi-terminal HVDC networks.                       |

TABLE 4. Summary of methodologies and key insights in underground cable fault detection.

| S.No | Methodology Focus                                                 | Common Themes & Techniques                                                                             | Key Insights                                                                                                       |
|------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1    | Ohm’s Law with Hardware (Micro-controller, Arduino)               | Simple, practical implementations for fault distance/location [106], [107], [108], [109], [110], [118] | Widely used for basic fault detection; suitable for educational & low-cost solutions but limited accuracy.         |
| 2    | Wavelet Transform & ANFIS (Adaptive Neuro-Fuzzy Inference System) | Signal processing with AI [111], [117], [119], [126], [127]                                            | High accuracy fault classification and location; handles noise well; often paired with discrete wavelet transform. |
| 3    | Deep Learning (LSTM, CNN, SVM)                                    | Modern AI for fault classification & localization [114], [120], [122]                                  | Achieved state-of-the-art accuracy (up to 99.6%); suitable for real-time, complex fault patterns.                  |
| 4    | Traveling Wave Analysis & Transient Modeling                      | Fast detection using physical wave properties [114], [115], [122]                                      | Fault detection in milliseconds; requires specialized sensors and transient analysis tools.                        |
| 5    | Impedance & Distance Relay Methods                                | Electrical parameter based fault detection [124], [125]                                                | Useful for incipient fault detection; complementary to signal analysis methods.                                    |
| 6    | MATLAB Modeling & Simulation                                      | Modeling cable parameters and behavior [121]                                                           | Essential for theoretical understanding and algorithm development before hardware implementation.                  |
| 7    | IoT & Data Logger Integration                                     | Smart grid and remote monitoring applications [109], [113], [127]                                      | Enables remote, real-time fault monitoring; relies on advanced communication and data analytics.                   |

- **Disadvantages:** The cable has not been easy for fault detection, has had high initial cost, and has required insulation with high dielectric strength.

6) SHEATH CABLE PARAMETER CALCULATION

The sheath resistance has depended on the resistivity of the material, cable length, and cross-sectional area. Sheath



**TABLE 5. Transient-based underground cable fault detection summary table.**

| S.No | citation | Introduction                                                                                    | Methodology                                                                                 | Conclusion                                                                                            | Accuracy                                                                                                                                                                                    |
|------|----------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | [139]    | Developed U-Cable model using frequency-dependent parameters and Laplace domain transformation. | Compared simulations using NLT and PSCAD tools.                                             | Differences were “barely perceptible.”                                                                | Solve time-domain equations via auxiliary differential method; validated with NLT and PSCAD; use dual Norton equivalent for simulation.                                                     |
| 2    | [140]    | Simulation-based study on overvoltage/inrush in PV solar plant cables; analyzed ferroresonance  | fault impact.                                                                               | Real fault scenarios in long underground cables in PV plants; validated using field and modeled data. | Error 0.2% at 1.8 km, up to 5% at 5.4 km length. Use IGBT modeling, surge arrester configuration, ampacity correction, and harmonic analysis; model the second and fifth harmonic behavior. |
| 3    | [141]    | Implemented RTWB wideband cable model with internal tap point access for real-time simulation.  | HYPERSIM platform; evaluated using two-segment (tapped) cable for computational efficiency. | 25% reduction in matrix-vector multiplications.                                                       | Reformulate admittance and propagation matrices to allow midpoint fault simulations with fewer computations; use tapped cable modeling.                                                     |
| 4    | [142]    | Built MATLAB GUI software for cable parameter estimation using the Carson Line Method.          | Validated with different cable types and cross-section modeling.                            | Impedance matrix accuracy depends on input data.                                                      | Use formulas for concentric neutral and tape shield cable; compute impedance matrix via Carson’s method; enable user selection of cable configurations.                                     |

**TABLE 6. AI-based underground cable fault detection summary table.**

| S.No | Title | Introduction                        | Methodology                                                         | Accuracy                              | Conclusion                                                                                 |
|------|-------|-------------------------------------|---------------------------------------------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------|
| 1    | [153] | ANFIS + DWT                         | Simulated fault conditions with varying location, resistance, angle | More than 99% classification rate     | Extracted wavelet energy, classified fault type, then locate using separate ANFIS networks |
| 2    | [154] | ANN + CWT in MATLAB Simulink        | IEEE 13-bus system; LG, LLG, LLLG faults                            | Not numerically specified             | Use time delay in current/voltage signals, processed by ANN for fault location             |
| 3    | [155] | CNN-LSTM                            | Simulated data with signal noise                                    | High accuracy; generalizable          | sliding window on voltage/current, trained CNN-LSTM, apply transfer learning               |
| 4    | [156] | LSTM + Resonance Frequency Analysis | MATLAB simulation with induced faults                               | Effective; precise fault localization | Detect resonance frequency shifts, train LSTM to learned frequency-location relation       |
| 5    | [157] | 1D-CNN + BSVM + FrDCT + SVD         | ATP/EMTP simulations; tested for real-time                          | 99.6% accuracy, 0.095% location error | FrDCT & SVD feature extracted, BSVM detected, CNN for location                             |
| 6    | [158] | ANFIS + DWT                         | ATP/EMTP simulation                                                 | <3% error in location                 | Current signal → DWT → feature extracted → ANFIS training & classification                 |
| 7    | [159] | PNN + DWT                           | Simulation & real signal                                            | High; accurate fault classification   | DWT extracted features → PNN classified faults across multiple system setups               |

inductance and capacitance have depended on the logarithmic values of self and mutual inductance. The sheath voltage has been calculated using the imaginary component of the product of angular frequency, mutual coupling between the core and sheath, and current flow [96].

## B. THERMAL CONSIDERATION

When a source has been applied, the cable has produced Joule heating ( $I^2R$  losses). The thermal resistance ( $R_t$ ) has

been determined by the properties of soil and insulation. The rated current has been maintained to avoid insulation breakdown [99].

## C. MECHANICAL CONSIDERATION

The cable bending radius has depended on mechanical stress, which may damage the insulation. The tensile strength has indicated the maximum force the cable can withstand during installation [100].

TABLE 7. Search strategy: queries used in the current study.

| S. No. | Search Query String                                                    |
|--------|------------------------------------------------------------------------|
| 1      | Underground cable AND incipient fault                                  |
| 2      | Underground cable AND arcing fault                                     |
| 3      | Underground cable AND incipient AND ML                                 |
| 4      | Underground cable AND incipient AND DL                                 |
| 5      | Underground cable AND incipient AND Arduino                            |
| 6      | Underground cable AND incipient AND IoT                                |
| 7      | Underground cable AND incipient AND ML OR DL                           |
| 8      | Underground cable AND incipient AND Ohm's Law<br>OR Travelling Wave    |
| 9      | Underground cable AND incipient AND Arduino OR<br>IoT                  |
| 10     | Underground cable AND incipient AND Varley<br>Bridge AND Moorey Bridge |

D. INSULATION CONSIDERATION

The capacitance of insulation has been determined by its relative permittivity [101]. The dielectric loss factor has represented the power loss within the insulation. During fault duration, sheath and armor resistances have been high, making proper grounding essential [102].

E. ENVIRONMENTAL CONSIDERATIONS

Soil thermal resistivity has influenced heat dissipation. The burial depth has varied based on thermal and electrical property changes, as well as moisture content in the soil [103].

IV. FAULT

An underground cable is widely used in urban/metropolitan areas. Mostly the cable experiences incipient and arcing faults in medium voltage distribution. The incipient fault is called a self-clearing fault for a certain duration turning into a permanent fault [104], [105], [106], [107], [108]. In traditional, 10-15 percent of cable faults proceed by incipient fault [110]. Over the period of incipient fault research on multicycle or sub-cycle detection in self-clearance fault of disturbance for than one cycle, it may be a disturbance lasting 1-4 cycles [104]. Many methods have been proposed for arcing fault detection [116]. The proposed methods are high impedance fault detection [112], ohm's law, Time domain reflectometry [117], fast Fourier transform analysis, wavelet transform [117], sliding window [118], tracing method, etc. In the kizhicay's arc fault analysis for cable voltage was a moderate value [113], [114]. The shape of a rectangular voltage waveform can be modeled by using Time Domain Reflectometry and Wavelet Transform. THD is used to find the fault location [115]. If the water penetrates in the cable for the reason of failure of insulation. It will generate loss in the form of water and electrical trees. Both trees are divergent of electric stress into the cable to discharge some voltage including trunks and branches. The water also called an electrical tree, which initiates 150 kV/cm [116], [124]. So the reason for the cable sheath randomly busts on the entire dielectric medium and breakdown occurs. Large water trees are calculated during over voltage caused

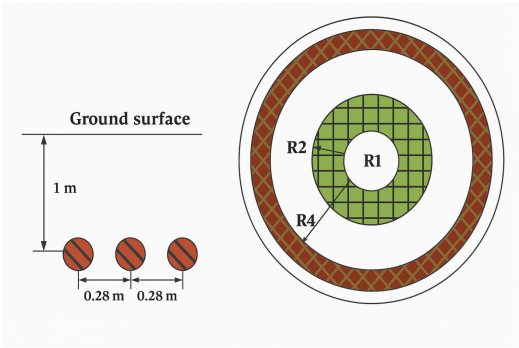


FIGURE 4. The configuration of cable in the simulation study [136].

TABLE 8. Radius measurements at different multiples of R [136].

| Multiples | Radius (mm) |
|-----------|-------------|
| $R_1$     | 0.00        |
| $R_2$     | 6.75        |
| $R_3$     | 10.15       |
| $R_4$     | 12.20       |
| $R_5$     | 13.80       |

by switching transients, the same as lightning. Numerous methods have “trees” as a result of electric stress divergence within the insulation, discharging voltage in patterns that include trunk and branch-like structures. Water trees can also evolve into electrical trees, with initiation fields around 150 kV/cm [116], [124]. This often results in random dielectric breakdown, where the cable sheath ruptures, damaging the insulation. Large water treeing typically occurs during over voltage events, such as switching transients or lightning strikes, both of which accelerate insulation degradation. A Comprehensive summary of methodologies and principal insights in underground cable fault detection table: 4.

Numerous methods have been proposed for arcing fault detection, including:

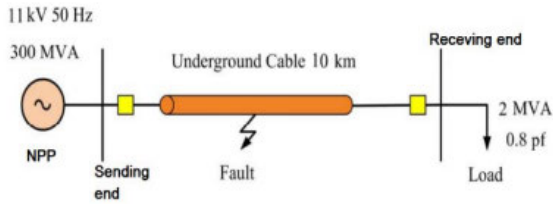
- High impedance fault detection
- Ohm's Law-based analysis
- Time Domain Reflectometry
- Fast Fourier Transform
- Wavelet Transform
- Sliding window techniques
- Tracing methods

If the occurrence of arcing faults in cables, the voltage profile exhibits a medium electric field and the result of the waveform is a rectangular waveform. This waveform can be analyzed using Total Harmonic Distortion (THD) with respect to arcing time to identify fault locations [104].

V. FAULT LOCATION

A. WAVE PROPAGATION

Wave propagation is used to locate faults into cables by sending electrical signals repeatedly in the form of high-frequency signal pulses or waveforms [30], [31], [32],



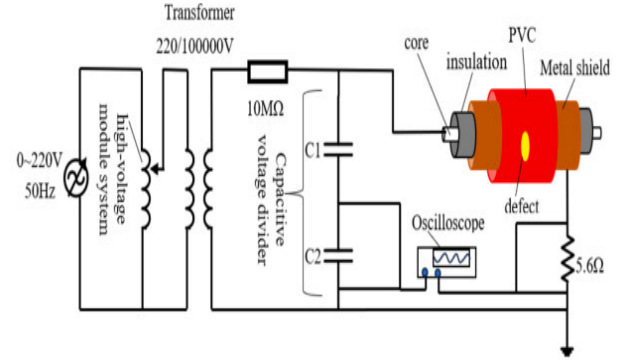
**FIGURE 5.** The system used in simulation studies, a 10-km three-phase Cross-Linked PolyEthylene (XLPE) stranded copper conductor single-core power cable, and the loading [136].

[33]. It throws the cable and analyzes how this waves behave, particularly how they reflect back from the fault. This method releases one principle of radar or sonar [34], [35]. It is known as time domain reflectometry (TDR) or pulse reflection or the sliding window technique. The system used in simulation studies, a 10-km three-phase Cross-Linked PolyEthylene (XLPE) stranded copper conductor single-core power cable, and the loading in the figure: 5. Inject the Gaussian pulse in the underground cable at one end to stimulate. It propagates through the cable, and the pulse is detected by analyzing the reflection and attenuation of the wave [36], [37]. Resistance and conductance are small changes caused by the fault. Reflection is discontinued, and then decide that the fault occurs in the cable. procedure: Signal Injection: TDR or pulse reflection or a sliding window device sends an impulse electrical pulse or waves down the cable [38], [39]. Wave Propagation: The wave travels through the cable. In a healthy cable, the signal travels without significant reflection [40], [41], [42]. Fault Reflection: When the wave enters a fault in cable (like a break, short, or impedance mismatch), part or all of the wave reflects back toward the source [43], [44], [45]. Time Measurement: The TDR or pulse reflection or sliding window measures the time it takes for the reflected wave to return [46], [47], [48]. Distance Calculation: Since the speed of wave propagation in the cable is known, the distance to the fault. The distance to a fault in a cable can also be determined using the propagation speed of the signal and the time it takes for the signal to reflect back from the fault point [49], [50], [51]. The travelling-wave phenomenon has been used to detect and locate underground cable faults by analyzing summary table: 1 The formula is given by:

$$\text{Fault Distance} = \frac{v \cdot t}{2} \quad (1)$$

$v$  = propagation velocity of the signal  
 $t$  = time taken for the signal to return

In the case of arcing faults in cables, the voltage profile exhibits a moderate electric field and typically resembles a rectangular waveform. This waveform can be analyzed using Total Harmonic Distortion (THD) to identify fault locations [52], [53], [54]. If water penetrates the cable, it can compromise the insulation, leading to losses and the formation of both water trees and electrical trees. This is the propagation velocity of the signal in the cable [55], [56], [57].



**FIGURE 6.** Measurement points and lab test setup [161].

The velocity  $v$  can be calculated using the inductance ( $L$ ) and capacitance ( $C$ ) per unit length of the cable as:

$$v = \frac{1}{\sqrt{LC}} \quad (2)$$

$L$  is the inductance per unit length (H/m),  $C$  is the capacitance per unit length (F/m).

#### 1) LIMITATIONS

- This method has been used for long cables where attenuation and dispersion have been more sensitive.
- This method has required high-speed sensors and precise time-stamping.
- Multiple reflections have complicated the interpretation.

#### 2) DEMERITS

- The equipment is expensive, making it unsuitable for low-budget utilities.
- These methods have not distinguished faults near cable ends easily.
- Jointing, bends, and insulation aging have influenced accuracy.

### B. TRANSIENT VOLTAGE CALCULATION

In the switching time, voltage transients and surges occur in the cable. It may influence with charging current effect and produce voltage stress on the insulation. The Feranti effect may increase the receiving voltage higher than the sending voltage. So the reason for harmonics is affected in the rectangular waveform [62], [63], [64], [65]. Transient voltage calculation is a powerful method for to find locating underground cable faults, Measurement points, and lab test setups shown in the figure: 6 especially when the fault causes a sudden change in electrical conditions, such as a short circuit or insulation breakdown [62], [66], [67]. A transient voltage is calculated as a short-duration surge or disturbance in voltage, reason by:

- Fault initiation like cable insulation failure
- Switching events
- Lightning strikes or sudden load changes



In underground cables, when a fault occurs, it repeats and causes a transient wave to propagate entire the cable [62], [67]. By analyzing these transient waves, the location of the fault can be estimated. Procedure: Fault Initiation: When a fault occurs phase-to-earth or phase-to-phase a transient voltage is generated at the fault point [62]. Wave Propagation: This transient wave travels in both directions the cable from the fault point [30], [68], [69]. Wave Detection: Sensors like voltage dividers, capacitive couplers, or current transformers are installed at one or both ends of the cable detect the arriving wave [70]. Time difference calculation: The Knowledge is the key to success of the transient wave is measured [71], [72]. Transient-based underground cable fault detection summary table: 5 Types of Transient-Based Fault Location Methods.

Single-Ended Method: Uses a sensor at one end of the cable. Relies on reflection of the transient wave [73], [74]. Double-Ended Method: Uses sensors at both ends. Measures the time of Knowledge is the key to success to locate the fault more accurately [75]. Algorithms have analyzed distorted current/voltage waveforms that have appeared during incipient faults summary table: 2. The voltage response during a transient event, such as a traveling wave due to a fault, can be expressed as a damped oscillation:

$$V(t) = V_0 \cdot e^{-\alpha t} \cdot \cos(\omega_d t + \phi) \quad (3)$$

$V_0$  is the initial amplitude, where  $\alpha$  is the damping coefficient,  $\omega_d$  is the damped angular frequency, and  $\phi$  is the phase angle.

The surge impedance (also known as the characteristic impedance) of a cable is given by: where:

$$Z = \sqrt{\frac{L}{C}} \quad (4)$$

$L$  is the inductance per unit length (H/m),  $C$  is the capacitance per unit length (F/m).

Solving the distance to the fault  $x$ : where:

$l$  is the length of each cable segment,

$r$  is the resistance ratio  $RA/RB$

$x$  is the distance to the fault from the test end.

Limitations: Assumes the presence of only a single fault.

Less sensitive to high-resistance faults.

## 1) LIMITATIONS

- This method specializes in sensors for capturing fast transients.
- Transients have been masked by switching operations or harmonics.
- Multiple reflections have made the interpretation complex.
- This method has not detected high-impedance faults effectively because they have not produced significant impedance changes.
- This method has been affected by load variations, which have influenced the measured impedance.
- This method has been difficult to apply in branched network configurations.

## 2) DEMERITS

- Installation and maintenance have been expensive.
- Application in branched networks has been difficult.
- Noise has been misinterpreted as fault-related transients.
- This method has required precise modeling of cable parameters.
- Temperature and soil moisture variations have affected the accuracy of this method.
- This method is less accurate for short or intermittent faults.

## C. WAVELET TRANSFORM

Wavelet Transform for fault detection in underground cables is one of the versatile method, especially for analyzing switching transients and non-stationary signals that were generated during faults [58], [59]. The cable breakdown for understanding and calculating fault detection using wavelet transform. Look for sudden spikes in the coefficients ( $D1$ ,  $D2...$ ), which indicate a transient or fault [60], [61]. Fault detection techniques using wavelets and neural networks-based underground cable fault detection summary table: 3. The procedure for incipient fault location includes:

- Fault time detection
- Incipient fault voltage phase angle verification
- Fault point voltage calculation
- Fault point voltage THD calculation
- Threshold comparison
- Load loss checking.

## 1) LIMITATIONS

- This method depended on sampling frequency and noise.
- This method required accurate synchronization for transient analysis.
- This method has high computational complexity in real-time applications.

## 2) DEMERITS

- No universal wavelet has worked for all cases, and the choice of wavelet has influenced accuracy.
- This method has been vulnerable to distortion caused by loads, harmonics, and switching operations.
- This method has required advanced measurement hardware.

## D. MURRAY LOOP METHOD

The Murray Loop Test is a classic and widely set method to locate faults in underground cables, especially when there is one faulty conductor and another healthy conductor available [21]. It's particularly effective for low-resistance faults include short circuits or earth faults [22]. Working Principle The Murray Loop Test is based on the principle of a Wheatstone bridge. It measures the resistance in the faulty conductor relative to a healthy conductor, this cable assuming the resistance is proportional to the cable length [23], [24].

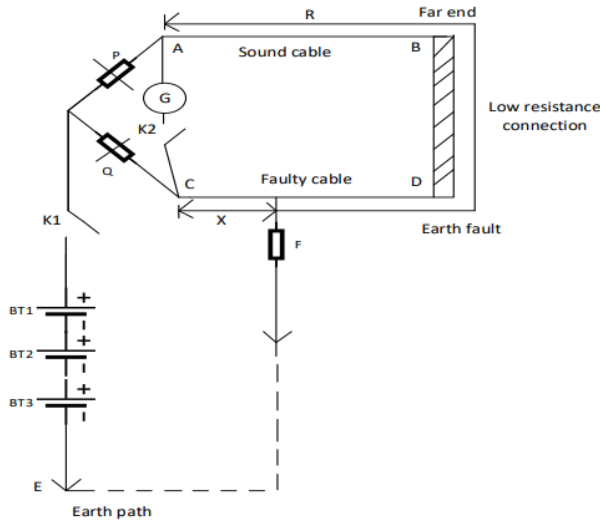


FIGURE 7. Murray loop method [157].

It requires a sound (fault-free) cable connected in parallel with the faulty cable [25], [26]. The Murray loop method is used to find fault location by requiring and employing a sound cable along the line using the Wheatstone bridge principle method, as illustrated in the figure: 7. Assumes uniform resistance per unit length along the cable [22], [27].

Requirements:

- One faulty conductor with a short or ground fault.
- One healthy conductor same type and length.
- Access to both ends of the cable.
- A galvanometer or null detector.
- A regulated DC power supply.
- Known resistance arms with variable resistors or known values.

#### Circuit diagram:

1. Connect the fault and sound conductors at the remote end..
2. At the test end, connect the conductors to form a Wheatstone bridge:

- One side of the bridge includes variable resistors R1 and R2.
- The other side consists of the faulty and healthy conductor loop.
- The DC supply and galvanometer are connected across the bridge.
- Fault must be of low resistance relative to the cable resistance.

In the Murray loop method was illustrated in the figure: 8, sound and fault cables are short-resistances Pair of resistances across the voltage source was tested. In the bridge circuited galvanometer connected between the resistor and test end. By adjusting the resistance to balance the galvanometer to the null position for calculation of location. However, the limitation of this method is less sensitivity to high fault resistance [28], [29]. Let  $R_C$  be proportional to  $1 + (lx)$  and

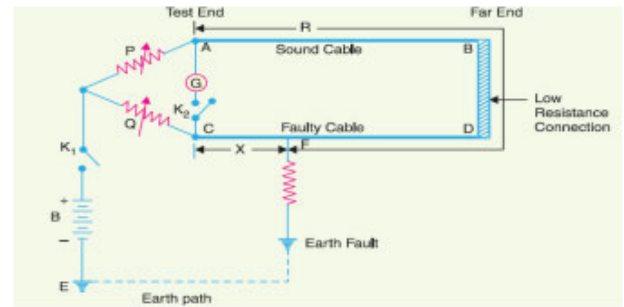


FIGURE 8. Murray loop [160].

RD be proportional to  $1$ . The bridge balance condition as expressed as

$$\frac{R_A}{R_B} = r = \frac{R_C}{R_D} = \frac{2l - x}{x} \quad (5)$$

Solving for the distance to the fault  $x$ :

$$x = \frac{2l}{r + 1} \quad (6)$$

where:

- $l$  is the length of each cable segment,
- $r$  is the resistance ratio  $R_A/R_B$ ,
- $x$  is the distance to the fault from the test end.

#### 1) LIMITATIONS

- Murray method required at least one healthy core of equal length.
- This method has assumed a uniform resistance per unit length, which is not always true.
- This method has shown less accuracy in detecting high-impedance faults.

#### 2) DEMERITS

- This method has not located multiple simultaneous faults.
- Soil conditions and moisture contact have affected its accuracy.
- This method has not been suitable for long cables with high attenuation.

#### E. VARLEY LOOP METHOD

The Varley Loop Method is a classical technique used to locate faults in underground cables, especially earth (ground) faults and short-circuit faults [76], [77]. This method was illustrated in the figure: 9. This method based on the principles of the Wheatstone Bridge and is especially suitable for low-resistance faults. If the fault resistance is high, the sensitivity of the Murray bridge method is reduced [78]. In such cases, the Varley loop method may be more suitable for earth fault figure: 10 and short circuit fault figure: 11 but it assumes that only a single fault exists [79], [80].

In contrast to the Murray loop method, where the ratio arms are variable, the Varley loop has fixed ratio arms [81], [82]. A variable resistance is inserted at the test end of the faulty

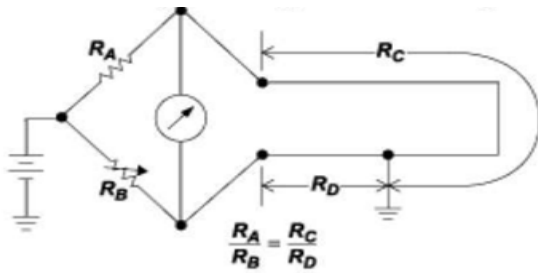


FIGURE 9. Varley loop [127].

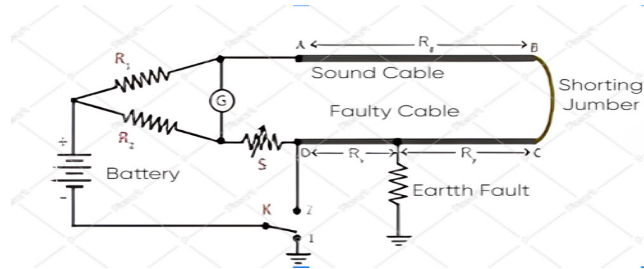


FIGURE 10. Varley loop method for earthed fault [127].

cable, improving sensitivity compared to the Murray loop method, especially when the fault resistance is high [83].

Locating a short circuit in the underground fault cable by using a Wheatstone bridge for small changes in the circuit.  $R_1$  and  $R_2$  are fixed, variable inserted leg in the bridge circuit, fault resistance ( $R_f$ ) is high. Sensitivity is better than the Murray loop method. The cable and the faulty sound cable are short-circuited at one end. A Faulty cable was connected at the first position to the negative to the battery. Connected to the cable ground in between two solid conductors. Unknown resistance can find the  $R_x$  and  $R_y$  is found by Wheatstone resistance calculation [84].

#### Setup Requirements:

- One faulty conductor with a known fault to ground or another conductor.
- Another sound (healthy) conductor to form the loop.
- A variable resistor.
- A galvanometer or null detector.
- A DC power supply.

#### Known cable length and resistance per unit length. Steps:

- Connection as per the circuit diagram. the faulty conductor and a healthy conductor to make a loop.
- Connect the loop to a bridge circuit like Wheatstone.
- Insert a variable resistor ( $R$ ) in series with the faulty leg.
- Adjust  $R$  through the galvanometer no deflection, indicating the bridge is balanced.
- Measure the value of  $R$  at balance.
- Use the known total resistance to calculate the distance to the fault, based on the cable's resistance per unit length and total length [84].

The equation relating the resistances is given by:

$$\frac{R_1}{R_2} = \frac{R_g + R_y}{R_x + S_1} \quad (7)$$

where:

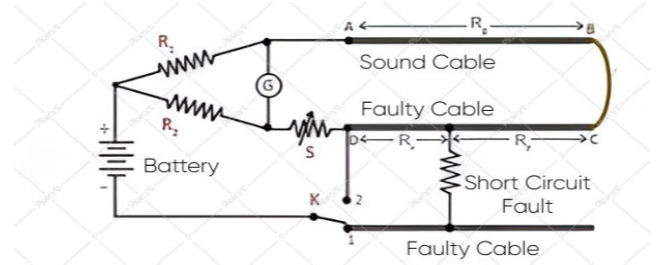


FIGURE 11. Varley loop method for short circuit fault [127].

- $R_1$  and  $R_2$  are the fixed resistances in the Wheatstone bridge,
- $R_g$  is the known resistance in the circuit,
- $R_x$  and  $R_y$  are the unknown resistances to be determined?
- $S_1$  is a small adjusting resistance,
- $R_f$  is the fault resistance.

Additionally, the sum of the resistances can be expressed as:

$$R_g + R_y = R_3 \quad (8)$$

#### F. OHMS LAW WITH IoT

Ohm's Law is a basic principle used in many electrical fault detection methods, including underground cable fault location [85], [86]. While Ohm's Law by itself the voltage is directly proportional to current and resistance. Resistance is directly proportional to the resistivity, length of conductor and indirectly proportional to the cross sectional area of conductor [87], [88]. When the fault occurs underground cable having value of resistance changes. So In this paper estimate the location of the fault [89], [90].

#### Steps in the location of the fault using Ohm's law:

##### 1. Known Cable Parameters:

The resistance per unit length of the cable (e.g., ohms/km). The total length of the cable.

##### 2. Inject a Test Voltage:

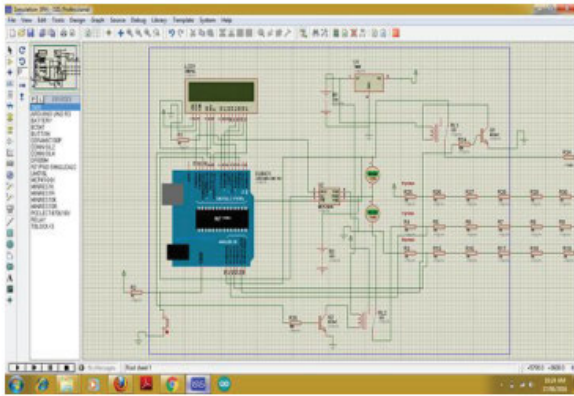
To apply a known DC voltage at one end of the cable.

##### 3. Measure Current or Resistance:

Measure the voltage and current or just the resistance from the end of the cable to the fault point using instruments like a megger or cable fault locator [83]. The underground cable fault simulation using Ohm's law method with Arduino has been carried out, as illustrated in the figure: 12. It was used to determine the appropriate optimization technique, an AI-based summary table: 6 for Underground Cable Fault Detection has been developed.

#### 1) LIMITATIONS

- The accuracy of long-distance cables has decreased due to distributed parameters.
- The performance has deteriorated when the fault resistance has been high.
- Some methods have required access to both ends of the cable.



**FIGURE 12.** Underground cable fault location by using arduino and linear variable resistor [21].

- The resistance measurements have been affected by sensitivity to temperature variations.
- Network latency has caused detection delays.
- Systems have depended on continuous connectivity and a stable power supply.
- This method has been affected by sensor calibration issues.

## 2) DEMERITS

- These methods have been suitable mainly for low-cost, low-voltage systems.
- This method has not detected incipient, arcing, or transient faults accurately.
- This method has not been reliable in multi-core or multi-branch networks.
- This method has been vulnerable to cybersecurity threats.
- The complexity of multi-device communication systems has increased.
- The implementation cost for large-scale deployment has been high.

## VI. FUTURE RESEARCH CHALLENGES

### A. HYBRID DEEP LEARNING MODELS

Future work has to integrate hybrid deep learning models such as CNN-LSTM, ANFIS-DWT, and transformer networks for real-time fault location models, which have enhanced the detection of:

- multicycle incipient faults,
- high-impedance faults,
- intermittent arcing faults.

The above methods are to improve accuracy under noisy and dynamic loading conditions.

### B. DATASET LIMITATIONS AND STANDARDIZATION

The absence of large, labeled underground cable datasets were the limitation. Research efforts have focused on:

- open-source dataset creation,

- synthetic dataset generation using EMTP/ATP/PSCAD,
- dataset standardization for ML/DL benchmarking.

### C. HIGH-RESOLUTION SENSING FOR TRANSIENT DETECTION

Wavelet and traveling-wave methods have required high sampling rates for accurate transient detection. Studies have explored:

- ultra-wideband sensors,
- fiber-optic distributed sensing,
- synchronized PMUs.

These technologies have enabled millisecond-level fault detection.

### D. IoT AND EDGE COMPUTING

IoT-based systems have suffered from latency in fault location processes, which has shifted computation toward:

- edge processors,
- embedded AI chips,
- fog computing.

### E. ADVANCED MODELING OF WATER AND ELECTRICAL TREES

dielectric degradation research models has developed:

- physics-informed ML techniques,
- improved dielectric degradation simulations,
- correlations between water-tree growth and fault initiation.

### F. FAULT LOCATION IN BRANCHED AND MULTI-TERMINAL NETWORKS

Traditional FL methods have assumed straight cable paths. New research has targeted:

- branched MV-LV hybrid networks,
- multi-ended synchronized measurement,
- sheath-current-based fault localization.

### G. ML-ASSISTED OPTIMIZATION OF CABLE PARAMETERS

Optimization studies have used:

- genetic algorithms,
- particle swarm optimization (PSO),
- grey wolf optimizer,
- spider-web-inspired optimization (SWO).

### H. NOISE, HARMONIC, AND LOAD ROBUSTNESS

Future algorithms have been required to be:

- noise-resistant,
- harmonic-resistant,
- load-independent,

### I. INTEGRATION WITH DIGITAL TWIN ENVIRONMENTS

Digital twins have enabled:

- real-time fault simulation,
- predictive maintenance,



- automated FLISR.

Future work has integrated ML with PMU, SCADA, AMI, and GIS systems.

## J. STANDARDIZATION AND FIELD VALIDATION

Future work has required to simulation-based on Standardization and Field Validation:

- field trials,
- testing under varying soil conditions,
- standardized fault-location test procedures.

## VII. CONCLUSION

This systematic review work presents underground cable modeling to find fault location techniques using various methodologies. Demonstration of cable modeling, fault type, and fault location significant progress, challenges. To overcome these challenges, future research should focus on improving the accuracy of finding exact estimation location techniques. Collaboration among the researcher, underground cable professional, and industry to find fault location. This approach could enable accurate fault location in the future.

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